AI461 ADVANCED BIOMETRIC SYSTEMS & SECURITY • COURSE PROJECT

Learning a Fixed-Length Fingerprint Representation

Implementation, Replication & Comparative Analysis of DeepPrint (IEEE TPAMI 2021)

Reference: Joshua J. Engelsma, Kai Cao, and Anil K. Jain

Traditional AFIS Challenge

Variable-length minutiae sets require expensive $O(N^2)$ graph matching.

DeepPrint Solution

Maps prints to a 192-D unit hypersphere vector for O(d) cosine matching.

Our Project Scope

Benchmarked ResNet-50 Baseline against full Dual-Branch STN on SOCOFing.

200-Byte Compact Template • ~1.25 µs Verification Speed

BENCHMARK DATASET

SOCOFing Dataset & Zero-Leakage Protocol

Pristine Dataset Scale

- **6,000 Real Prints:** 600 subjects × 10 fingers.
- **Pristine Gallery:** Unaltered AFIS enrollment prints.
- **Sensor Resolution:** Standardized 500 DPI optical capture.

Synthetic Alteration Tiers

- **Easy (17,931):** Mild moisture and surface smudges.
- **Medium (17,067):** Noticeable cuts and tissue scars.
- **Hard (14,272):** Severe core obliteration and deep cuts.

Distortion Mechanisms

- **Central Rotation (CR):** Arbitrary scanner misalignment.
- **Obliteration (Obl):** Burns and abrasions erasing ridges.
- **Z-Cut Scars (Zcut):** Severe surgical ridge breaks.

Strict Subject-Independent Open-Set Protocol (Zero Data Leakage)

Training: 70 Subjects (700 fingers) • **Held-Out Testing:** 15 Unseen Subjects (150 fingers, 1,250 queries)

1,250 Queries
100% Unseen Identities

MODEL ARCHITECTURES

Baseline ResNet-50 vs. DeepPrint Dual-Branch

1. Baseline ResNet-50 Architecture

224×224 CLAHE → ResNet-50 → GAP (2048-D) → FC (192-D) → L2 Sphere

- **Backbone:** Pretrained ResNet-50 conv stages + BatchNorm1D.
- **No Alignment:** Fails severely when fingerprint is rotated.
- **Global Texture Only:** Susceptible to localized Z-cut scars.
- **No Minutiae Prior:** Lacks inductive bias for ridge endings.

2. DeepPrint Paper Dual-Branch Architecture

Input → STN Alignment (2×3 Affine) → Shared Stem → Dual Branches → 192-D Vector

- **Spatial Transformer (STN):** Predicts 2×3 affine matrix θ for canonical alignment.
- **Texture Branch (96-D):** Captures global ridge flow and pattern frequencies.
- **Minutiae Branch (96-D):** Learns bifurcations via 6-ch heatmap decoder (training only).
- **L2 Fusion:** Concatenates [r_min, r_tex] into unit vector $R \in S^{191}$ (decoder discarded at test).

Reference: Engelsma, Cao & Jain, IEEE TPAMI 2021 (Figure 4 Architecture)

END-TO-END NETWORK TOPOLOGY

Implemented Dual-Branch Architecture Pipeline



1. Alignment & Shared Stem

Input (3 × 224 × 224) → STN Alignment → Shared Stem (Conv1 + BN + ReLU + MaxPool + Layer1).

2. Parallel Dual Branches (ResNet Layer2, Layer3, Layer4)

- **Texture Path:** Layer2–4 → GAP → **FC (512 → 96)** → 96-D Texture Embedding & Classifier.
- **Minutiae Path:** Layer2–4 → GAP → **FC (512 → 96)** → 96-D Minutiae Embedding & Classifier.

3. Auxiliary Minutiae Decoder (Training Supervision)

Branches off Minutiae Layer4 → **Minutiae Decoder → Predicted Heatmap (6 × 64 × 64)**.

4. Feature Fusion & Hypersphere Normalization

Concatenate (96-D + 96-D) → 192-D Embedding → L2 Normalization → Final Representation.

THEORETICAL FORMULATION

Loss Functions & Mathematical Modeling

1. Multi-Task Objective (Section 3.4)

Joint identity cross-entropy for Texture & Minutiae heads + auxiliary heatmap MSE (λ = 0.1).

$$L_{\text{total}} = L_{\text{CE}}^{\text{tex}} + L_{\text{CE}}^{\text{min}} + \lambda \cdot L_{\text{MSE}}(\hat{H}, H)$$

2. 6-Ch Minutiae Heatmap (Section 3.3)

Quantizes [0, 2π) into K=6 bins (θ_k = kπ/3) smoothed via spatial (σ_s = 2.0) & angular kernels.

$$H(i,j,k) = \sum_t \exp\left(-\frac{d^2}{2\sigma_s^2}\right)\exp\left(-\frac{d_\phi(\theta_t,\theta_k)}{2\sigma_o^2}\right)$$

3. STN Affine Grid Sampler (Section 3.2)

Differentiable coordinate sampling learns end-to-end rotation and translation alignment.

$$[x_i^s, y_i^s]^T = A_\theta \cdot [x_i^t, y_i^t, 1]^T$$

ARCHITECTURAL ANALYSIS

Why the Dual-Branch Architecture is Superior

1. Eliminates Rotation Sensitivity

STN dynamically aligns rotated inputs, boosting Central Rotation (CR) Rank-1 from **85.95% to 99.05% (+13.10%)**.

2. Minutiae Inductive Bias

Auxiliary 6-channel heatmap loss prevents CNN shortcut learning on skin moisture or background noise.

3. Decoupled Representations

Separating macro Texture (96-D) from local Minutiae (96-D) stops localized Z-cut scars from corrupting the full vector.

4. Zero Runtime Overhead

The heatmap decoder is discarded at inference, keeping **200-byte** storage and **~1.25 µs** matching latency.

EMPIRICAL EVALUATION

Experimental Results on Held-Out Open Set

Dataset Condition	Baseline Rank-1	Dual-Branch Rank-1	Baseline TAR@0.1%	Dual-Branch TAR@0.1%
Altered-Easy	100.0%	**100.0%**	97.3%	**100.0% (+2.7%)**
Altered-Medium	96.1%	**100.0%**	78.6%	**96.8% (+18.2%)**
Altered-Hard	**83.3%**	**98.9%**	**50.0%**	**90.4% (+40.4%)**
Combined	93.8%	**99.7%**	75.4%	**95.5% (+20.2%)**
Compressed (200B)	93.8%	**99.7%**	75.4%	**95.5% (+20.2%)**
Equal Error Rate (EER)	**2.52%**	**0.60%**	—	**4.2× Error Reduction**

Evaluated on 15 Held-Out Subjects (150 Unseen Fingers, 1,250 Altered Queries)

ENGINEERING & DEPLOYMENT

200-Byte Compression & Microsecond Matching

200 Bytes
73.96% Smaller than 768B Float

0.999983
Cosine Correlation (MAE: 0.000364)

~1.25 µs
>750,000 Matches/Sec per Core

5M / 1 GB
Templates Stored in 1 GB RAM

Section 3.5: Min-Max Scalar Quantization

$$R_i' = \text{round}(255 \times \frac{R_i - \min(R)}{\max(R) - \min(R)})$$

192 uint8 bytes + 8 bytes (two float32 scalars) = **200 Bytes**. 1:N search (N=150) takes **0.015 ms**.

Live Match Inspector (Impostor Test)

- **Baseline False Match:** Cosine score **0.5747 > 0.45** threshold.
- **Dual-Branch True Reject:** Score drops to **0.0227 << 0.45**.

SUMMARY & CONCLUSION

Key Takeaways & Project Deliverables

1. Scientific Insights

- **Hard Alterations:** TAR@0.1% jumped from 50.0% to **90.4% (+40.4%)**.
- **Rotation Solved:** CR Rank-1 reached **99.05% (+13.1%)** via STN.
- **EER Drop:** Reduced 4.2× from 2.52% down to **0.60%**.

2. System Efficiency

- **Backbone Speed:** Trained ~4× faster with 50% fewer parameters.
- **Lossless 200B:** 0.999983 cosine correlation after quantization.
- **Throughput:** ~1.25 µs per match (750K matches/sec/core).

3. Project Deliverables

- **Training Notebook:** Full STN + Dual-Branch PyTorch suite.
- **Evaluation Engine:** Automated open-set benchmark in `test.py`.
- **AFIS Web Portal:** Interactive match inspector in `frontend/index.html`.

Thank You! Questions & Discussion

Live AFIS Dashboard Demonstration Ready • frontend/index.html

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